

TYLER VINCENT

tylervincent@alumni.usc.edu • Los Angeles, CA • tylerv11.github.io

SUMMARY

Data Engineer (Analytics/BI) with 5+ years building governed pipelines and BI products at enterprise scale - Databricks, Python, SQL. Lean Six Sigma Black Belt with a track record of turning operational data into actionable decisions that move the business.

EXPERIENCE

TAKEDA PHARMACEUTICALS

Los Angeles, CA

Sr Data & Analytics Engineer

Aug 2023 - Present

- Eliminated 15,000+ hrs/yr of manual compliance reconciliation across 23+ global sites by engineering a production-grade Databricks silver-to-gold pipeline, processing 15M+ records with CTE-based logic, dynamic row-level security, and regional scoping for 1,500-2,000 MAU
- Quantified \$6.5M in manufacturing throughput opportunity, by translating event-based production data into digital value stream maps and manufacturing analytics for throughput, float, and process drift using validated Lean Six Sigma metrics (Cpk, float) in SQL, DAX, QlikScript
- Reduced cycle times 27% (30 to 22 min) by building a Flask-based Databricks writeback app to surface, Pareto, and target production delays, capturing technician delay inputs with prepopulated batch context, validated joins against master data, and writing records in Delta tables
- Established AI-ready data foundations by designing semantic-model instruction layers, synonym mappings, and Copilot guardrails serving hundreds of users daily; coached 400+ teammates through contextualizing AI outputs against validated enterprise datasets in group sessions
- Built and governed 40+ datasets, 25+ reports, and 17 reusable, modular Databricks pipelines, shifting analytics ownership to domain experts and enabling self-service deployment across manufacturing operations

CHEWY.COM

Boston, MA

Systems Engineer II

Oct 2021 - May 2023

- Identified \$700K in cost savings and \$750K in recoverable revenue from phone payment failures by conducting multivariate data analysis in Minitab on transaction patterns, then designed and validated process improvements through controlled A/B testing
- Drove 20% to 94% user adoption (netting \$400K savings) by analyzing agent click-level workflow data in Python, identifying friction points in the user journey, and iterating product design based on quantified engagement metrics
- Projected \$716K annualized savings and 1.6% handle-time reduction across 2M+ monthly contacts by executing two controlled studies on cycle time and first-contact resolution - analyzed results in Snowflake via custom SQL reports and secured executive buy-in for deployment
- Automated biweekly statistical testing across 1M+ incidents (normality, hypothesis, stability, regression) for executive board forecast reporting

GENERAL DYNAMICS ELECTRIC BOAT

North Kingstown, RI

Data & Process Engineer II · DoD Secret Clearance

Jun 2020 - Aug 2021

- Reduced transportation costs 26% by designing an inventory control and demand-planning database in SQL that linked legacy terminal-based material systems with shop-order data to improve equipment availability and staging decisions
- Cut facility setup time 54% and turnover time 21% across 250+ work centers by translating Gemba observations, VSM findings, and kanban constraints into Tableau and Excel operational dashboards consumed by production supervisors and engineering leads
- Digitized 200+ controlled work procedures into a permission-governed SharePoint platform spanning Secret, FOUO, and unclassified tiers, with AutoCAD layouts, manpower plans, and optimized staging schedules for 14,000 personnel

GRAYMATICS, INC. (APAC-focused SaaS video analytics startup)

Singapore

AI/ML Program Manager

Apr - Aug 2019

- Managed 7 cloud-to-edge computer vision deployments, including 2 product launches, across SE Asia; organized and led 20+ person international ML teams, generating \$100k+ in revenue by establishing product owners, scrum, and technical leads

NORTHERN TIER EXPEDITIONS

Thunder Bay, Canada

Senior Expedition Guide · Guided 14+ groups on 1,000mi int'l canoe treks

Summer 2017 - 18

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

Los Angeles, CA

B.S. Industrial & Systems Engineering · 4-year Dean's Scholarship (Top 2%) · Engineering and Writing Honors Program

National University of Singapore, International ISE Exchange Program (only USC student accepted) 2019

SKILLS AND CERTIFICATIONS

Skills: SQL · Python · Flask · Databricks · Delta Lake · Power BI · ETL/ELT · Star schema modeling · Data governance · Azure · Snowflake · Value stream mapping · Lean Six Sigma · SPC/variation analysis · A/B testing · Tableau · QlikSense · Jira · Scrum · HTML

Certifications: Lean Six Sigma Black Belt · Model Context Protocol · Agile PM (Microsoft Project)

PROJECTS

THE AEROSPACE CORPORATION *Resource Allocation Scheduler* (Senior Thesis, **Engineering Faculty Award**)

- Developed a Python multi-objective, optimization algorithm, scheduling 50+ military antennas, verifying and recording 100+ satellites and deep space objects in Geosynchronous Earth orbit with radar every 24 hours

Personal Multi-Agent AI Infrastructure (Hub-and-spoke orchestration, MCP architecture)

- Designed and deployed a local multi-agent LLM orchestration system using Model Context Protocol, routing research and analytics tasks across six models with sandboxed tool servers, approval gates, partitioned memory, audit logging, and zero-sudo OS access.